

Voice and Gesture Based Virtual Desktop Assistant for Physically Challenged People

SEMINAR REPORT

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the seminar report entitled “**Voice and Gesture Based Virtual Desktop Assistant for Physically Challenged People**” submitted by **Adil Mohammed Shajahan, SBC20CS002** in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Technology in Computer Science and Engineering** is a bonafide record of the work carried out under my guidance and supervision at college.

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ABSTRACT

Computers are getting better, faster, and smarter in the modern era. The fields of medicine, business administration, education, and other related fields are using them more and more. Therefore, it is necessary to make a computer's operability and usability simpler. It's well known that using digital virtual assistants makes interacting with computers easier. People who are deaf or hard of hearing find it challenging to use virtual assistants on their devices because the majority of digital virtual assistants rely on voice commands. The goal of this is to develop a voice and gesture-based virtual assistant that both disabled and non-disabled people can use to complete everyday computer tasks. The primary goal is to foster organic human-machine interaction. The user chooses what information to provide the virtual assistant. In addition to using voice and gestures for communication, users can also use gestures to move a mouse pointer. Gestures are identified by the suggested system even in dimly lit areas. Thus, in all weather conditions, the suggested system is quite useful. Being contact free is beneficial for both individuals with disabilities and during times such as the COVID-19 pandemic.

Keywords: HCI, ML

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ABBREVIATIONS

HCI	Human-Computer Interaction
ML	Machine Learning
Covid-19	Corona Virus Disease of 2019
MFCC	Mel-Frequency Cepstral Coefficients
HMM	Hidden Markov Model
NN	Neural Network
DNN	Deep Neural Network
SVM	Support Vector Machines
CNN	Convolutional Neural Network
NLP	Natural Language Processing
API	Application Interface
UI	User Interface
GUI	Graphical User Interface
VAcall	Virtual Assistant call
HGcall	Hand Gesture call
ANN	Artificial Neural Network
AR	Augmented Reality

CHAPTER 1

INTRODUCTION

1.1 Overview

A virtual assistant is a new kind of interactive technology that facilitates user interaction and helps with daily tasks. Applications like weather reports, shopping lists, reminders, and finding the right places or products are some examples. They are able to accept gesture, voice, and text commands. The desktop version of this system is intended for efficient use. Voice-activated virtual assistants are becoming more and more popular. Virtual assistants are currently freely accessible on the majority of gadgets. Virtual assistants are now more readily available, but their daily use is not as high as one might have thought, according to recent surveys. As a result, gestures are added to voice-based virtual assistants, which will be beneficial for those with physical disabilities. Additionally, finger-based mouse pointer control allows for contactless communication, which is particularly helpful during the Covid-19 pandemic. It is predicted that by 2024, there will be 8.4 billion virtual assistants in use worldwide, up from the 4.2 billion that existed in previous years [1], [2]. Virtual assistants are widely available in the market, and their network is growing in response to the features that they offer. Current System: While voice-based virtual assistants like Google, Siri, Alexa, and others are in use today, desktop software that combines voice and gesture-based virtual assistants is not currently feasible. A hardware device that was based on already published research was suggested, but it was never put into practice.

Even though voice-activated virtual assistants are doing a great job and making tasks easier, those who are deaf or dumb cannot use them. This is the goal, and gestures are used as a means of communication as part of the solution. This can be used more effectively and solve issues as image and video processing techniques advance.[2], [3].

1.2 System Model for Virtual Assistant

On our desktop computers, voice assistants can be used to speak commands to send emails, search the internet, receive news updates, navigate folders, open files and apps, and show schedules. Our desktops and laptops can be controlled by voice [4], [5]. Mouse pointer control based on gestures: A webcam records video, and you can move the mouse pointer, pick, or even select other operations with hand gestures.

Gesture-based task execution involves training a model to recognize specific hand gestures and carry out related tasks. The webcam streams live in order to record gestures. In the event that a gesture made by the user is recognized by the system as predefined, the associated task will be carried out [2], [4], [7].

Applications:

1. Humanization of robots: Humanoid robots are robots that interact just like humans. So there is a need for it to comprehend the spoken and gesture input from human beings and reply using speech just like humans.
2. Integration of digital assistants into a smart home system: The idea of smart home is to control appliances by communicating with them. Instead of conventional techniques like remote controls or switches, using voice is more convenient.
3. The transition from text to speech: Most people nowadays prefer listening to content than to read it. As part of digitalization books, articles, news etc are now available in digital format. Text to speech engines can be used to listen to text versions of content on the internet or even a document available offline. Using this feature people can gain information while doing some other tasks.
4. Personalization of voice assistant: Voice assistants store basic information about users. They respond and recommend in a way likable by users.
5. Conversational Commerce: As speech is a natural way of interacting, e-commerce websites are creating voice bots that can help users search for a product and purchase it.
6. It played a vital role in the fight against Covid-19: During the pandemic the use of digital assistants increased as people had to maintain social distance and to stay isolated.
7. Gesture based virtual assistant: It can be used for sign language based interaction with people who have problems with speech and hearing.
8. Gaming controls: Actions in a game can be controlled using voice based input.

1.2.1 Data Collection and Preprocessing

Voice Data Collection: Record a diverse range of spoken commands from a variety of individuals with different accents and speaking styles. Ensure the recording environment is quiet and free from background noise. **Gesture Data Collection:** Capture hand gestures using motion tracking devices or depth cameras. Collect gestures for various commands and actions, ensuring clear differentiation between gestures. **Data Preprocessing:** Apply noise reduction techniques to remove background noise from voice recordings. Normalize audio levels to

ensure consistent volume across recordings. Extract relevant features from voice data, such as pitch, formants, and mel-frequency cepstral coefficients (MFCCs). Extract features from gesture data, such as hand position, orientation, and motion trajectory.

1.2.2 Voice Recognition Model Training

Model Selection: Choose an appropriate voice recognition model architecture, such as Hidden Markov Models (HMMs), Neural Networks (NNs). **Model Training:** Train the selected model using the preprocessed voice data. Adjust model parameters and optimize the training process to achieve high accuracy in recognizing spoken words. **Model Evaluation:** Evaluate the performance of the trained model using a separate dataset of voice recordings.

1.2.3 Gesture Recognition Model Training

Model Selection: Select an appropriate gesture recognition model architecture, such as Support Vector Machines (SVMs). **Model Training:** Train the chosen model using the preprocessed gesture data. Optimize model parameters and training process to achieve high accuracy in identifying different hand gestures. **Model Evaluation:** Evaluate the performance of the trained model using a separate dataset of gesture recordings. Measure accuracy metrics such as classification accuracy and gesture recognition rate.

1.2.4 Natural Language Processing Integration

NLP Framework Selection: Choose an appropriate NLP framework, such as NLTK, spaCy, or TensorFlow NLP, to handle text processing tasks. **Intent Recognition:** Develop an intent recognition module within the NLP framework to classify text commands into different categories based on their intent. **Entity Extraction:** Implement an entity extraction module to extract relevant information from text commands, such as object names, locations, or time references.

1.2.5 Action Execution Module

Task Decomposition: Break down user requests into smaller, more manageable tasks. **Application Interface (API) Integration:** Integrate APIs for various applications and system

functions to enable the virtual assistant to perform tasks. Task Execution Logic: Develop logic to orchestrate the execution of tasks, including handling dependencies, error handling, and feedback mechanisms.

1.2.6 User Interface (UI) Design and Integration

User-Centered Design: Employ user-centered design principles to create an intuitive and user-friendly interface. Multimodal Input Handling: Design the UI to accept both voice and gesture-based inputs seamlessly. Visual Feedback: Provide visual feedback to users to confirm command recognition, task progress, and error handling. Component Integration: Integrate the individual components, including voice recognition, gesture recognition, NLP, action execution, and UI, into a unified system.

1.3 Methodology

The process of creating the virtual desktop assistant involved gathering a dataset of gesture and voice commands. This dataset was gathered from a group of individuals with physical disabilities who were asked to carry out a range of tasks, including using the media player, opening apps, and browsing the internet. A webcam and microphone were used to record the orders. The dataset was carefully selected to guarantee both its excellent quality and the inclusion of a representative sample of commands that individuals with physical disabilities might find useful. In order for the model to learn to associate the commands with the tasks they represent, the corresponding tasks were also labeled on the dataset. The gesture and speech data were processed to extract features. MFCCs serve as a representation of a sound's spectral content. Tasks involving speech recognition frequently make use of them. The hand gestures were used to represent the trajectory of the hand as it moved through the air, and position, orientation, and velocity were extracted from them. The machine learning model was able to comprehend the voice and gesture commands because of the features that were extracted. The voice and gesture commands from the dataset were recognized by the model. Neural networks with a spatial structure, such as CNNs, are particularly useful for extracting features from images and other data. Speech and gestures are examples of sequential data that are well suited for processing by RNNs, a kind of neural network. The performance of the model was evaluated on a held-out test set. The test set was a subset of the data that was not

used to train the model. This was done to ensure that the model was generalizable and could perform well on data that it had not seen before. Putting the model into practice in a virtual desktop assistant was the last stage. The assistant was made to be used by those with physical disabilities. It allows voice and gesture control over a computer. It might prove to be an invaluable resource for those with physical disabilities. It can make using computers easier and more autonomous for them.

1.4 Scheme of the Report

CHAPTER 1 Consists of an Introduction, Overview, and Methodology of technology.

CHAPTER 2 Consists of the reference papers of the seminar paper.

CHAPTER 3 Consists of system overview of the seminar paper.

CHAPTER 4 Consists of the advantages and disadvantages.

CHAPTER 5 Consists of future work

CHAPTER 6 Consists of conclusion of the seminar report.

CHAPTER 2

LITERATURE SURVEY

2.1 “A Vision and Speech Enabled, Customizable, Virtual Assistant for Smart Environments.” Iannizotto, Giancarlo, Lucia Io Bello, Andrea Nucita and Giorgio Mario Grasso. [1]

A type of AI assistant that can communicate with users and operate smart devices in their surroundings using both speech and vision is a virtual assistant for smart environments that is customizable and has vision and speech capabilities. Additionally, users can modify it to suit their unique requirements and tastes because it is highly customizable. Compared to traditional virtual assistants, which are usually restricted to speech-based interaction, this kind of virtual assistant offers a number of potential advantages. People with disabilities, such as those who are blind or have restricted mobility, can interact with their environment more easily with the assistance of a vision-enabled assistant. For routine tasks like playing music, adjusting the thermostat, and turning on lights, it can also be more practical and effective. A customizable virtual assistant allows users to tailor the assistant's appearance, voice, language, and personality to their own preferences. This can make the assistant more engaging and enjoyable to use. A specific architecture for a virtual assistant for smart environments that is customizable, vision and speech enabled. Because of its efficient and resource-efficient architecture, it can be used with inexpensive hardware like the Raspberry Pi. It also makes use of numerous cutting-edge methods in artificial intelligence, deep learning, computer vision, and speech generation and recognition. The results showed that the assistant was accurate, reliable, and appealing to users. All things considered, a speech- and vision-enabled, adaptable virtual assistant for smart environments could completely change how humans interact with our surroundings. It can improve the accessibility, efficiency, and convenience of our lives. People who are blind or have limited mobility can use the assistant to interact with their environment and control smart devices without having to get up and move around. The assistant can be used to perform a variety of tasks, such as turning on lights, adjusting the thermostat, and playing music, without having to use a separate device or app, monitor the security of a home or business or to check on the well-being of loved ones, control smart devices and access information even in areas where there is no internet access. Users can tailor the assistant to their specific needs and preferences, such as choosing a voice, language, and personality.

The drawback of this virtual assistant are;

Privacy concerns: The assistant may collect sensitive data about user activities and preferences. It is important to choose a virtual assistant from a reputable company with a strong privacy policy.

Security risks: The assistant may be vulnerable to hacking or other security threats. It is important to keep the assistant's software up to date and to use strong passwords.

Cost: Vision and speech enabled virtual assistants can be expensive to purchase and maintain.

Complexity: Vision and speech enabled virtual assistants can be complex to set up and use. It is important to choose an assistant that is easy to use and that has good customer support.

Limited functionality: Vision and speech enabled virtual assistants are still under development and may not be able to perform all of the tasks that you would like them to be able to do.

Dependence on technology: Vision and speech enabled virtual assistants are dependent on technology and may not work properly if there is a power outage or other technical problem.

Potential for misuse: Vision and speech enabled virtual assistants could be misused by malicious actors to spy on users or to spread misinformation.

2.2 “Real-Time Human Gesture Grading based on OpenPose” S. Qiao, Y. Wang and J.Li. [3]

Human gesture recognition is a challenging task due to the variability of human movements and the complexity of natural environments. Traditional methods for human gesture recognition often require expensive equipment, such as motion capture systems, or involve complex algorithms that are not suitable for real-time applications. Qiao, Wang, and Li's real-time human gesture grading system, which is based on OpenPose, uses a low-cost, real-time method to grade human gestures from monocular images in order to overcome these limitations. The system finds the 2D positions of an individual's joints in an image by using the OpenPose library, which offers real-time multi-person keypoint detection. The motion trajectory for each joint is then calculated using this information, and the gesture is then compared to a standard reference. The system consists of a monocular camera that captures an image of the person performing the gesture. The OpenPose library is used to detect the 2D positions of the person's joints in the image. The equation of motion trajectory is computed for each joint to capture the movement of the person's body. A similarity metric is defined to measure the distance between the motion trajectories of the standard and real-time videos.

This similarity metric represents the accuracy of the performed gesture. Modifiable scoring formula is used to simulate the gesture grading scenario. The grading algorithm takes into account the similarity metric and other factors, such as the importance of specific joints, to generate a final grade. The real-time human gesture grading system based on OpenPose has a wide range of potential applications including Sports Coaching, Rehabilitation, Entertainment, Dance Instruction as it gives the real-time feedback to the users regarding their posture, accuracy, improvement, etc. When compared to conventional techniques for human gesture recognition, the suggested system has the following advantages as Performance in real time: The system is appropriate for applications that need instant feedback because it can evaluate human gestures in real time. Low cost: The system's use of the OpenPose library, which is available under open source, lowers the total cost of implementation. Robustness to noise: The system can function in real-world settings because it is resilient to noise in the input image. Usefulness: With just a monocular camera and the OpenPose library needed, the system is comparatively simple to operate.

The drawbacks of gesture grading system based on OpenPose are;

Limited Gesture Complexity: The system may not be able to grade complex gestures accurately, especially those involving fine-grained movements. For example, it may be difficult to distinguish between different types of dance steps or to assess the technique of a gymnast performing a complex routine.

Training Data Dependency: The system requires training data to learn how to grade different gestures, which may be time-consuming and resource-intensive to collect. This can be a challenge for new gestures or for applications in domains where there is limited data available.

Environmental Limitations: The system's performance may be affected by lighting conditions, background clutter, or occlusions. For example, if a person's arm is occluded by another object, the system may not be able to detect the 2D positions of the joints correctly, which could lead to inaccurate grading.

Potential for Bias: The system may be biased towards certain types of gestures or towards the gestures that are included in the training data. This could lead to unfair or inaccurate grading, especially for gestures that are not well-represented in the training data.

Lack of Generalizability: The system may not be able to generalize well to new environments or to gestures that are not similar to the gestures that it was trained on. This could limit the system's applicability in real-world settings.

2.3 “Hand Gesture Vocalizer for Deaf and Dumb People”

Sangeethalakshmi K, K. G. Shanthi, A. Mohan Raj, S. Muthuselman, P. Mohammad Taha and S. Mohammed Shoaib. [14]

With the use of hand gesture vocalizers, deaf and mute people can communicate more successfully with non-sign language speakers by converting hand gestures into spoken words. Typically, they use a mix of text-to-speech synthesizers, machine learning algorithms, and sensors to accomplish this function. Wearable devices, like gloves, can have sensors attached to them, like accelerometers or flex sensors. The position and motion of the user's hand and fingers are recorded by these sensors. The central processing unit of the device is the microcontroller. It gathers, analyzes, and forwards sensor data to the machine learning algorithm for gesture recognition. The particular hand gesture being made is identified by the machine learning algorithm by analyzing the sensor data. The gesture is classified using a trained model that makes use of data-extracted patterns and features. The text-to-speech synthesizer turns the gesture into spoken language after it has been identified. The corresponding audio output is produced by means of a text-to-speech engine or prerecorded voice. Hand gesture vocalizers provide deaf and mute people with a number of benefits, such as: Enhanced Communication: They increase the number of communication options by facilitating communication with people who are not proficient in sign language. Increased Independence: By encouraging increased independence, they enable deaf and mute people to take part more fully in daily activities. Increased Self-Confidence: By promoting more efficient communication, they can increase social interaction and self-confidence. Some of the challenges include accuracy of hand gestures, cost of hand gesture vocalizer and some individual prefer sign language over the device. Hand gesture vocalizers have the potential to be an important assistive technology that will help the deaf and mute participate more fully in society and communicate more effectively. It is expected that technology will become more and more significant in helping deaf and mute people communicate with the hearing community as it develops and matures.

Hand gesture vocalizers offer several advantages for deaf and mute individuals, but they also face some drawbacks, some of drawbacks of hand gesture vocalizers are;

Accuracy: Current systems may not be able to accurately recognize all hand gestures, especially in noisy environments or when the user's hands are obscured which can lead to misunderstandings and frustration.

Cost: Hand gesture vocalizers can be expensive, which may limit their accessibility to some individuals i.e, cost of the sensors, microcontroller, machine learning algorithm, and text-to-speech synthesizer can add up quickly.

User Acceptance: Some deaf and mute individuals may prefer sign language as their primary mode of communication. They may feel more comfortable and confident using sign language, which is a natural language for them.

Social Integration: For hand gesture vocalizers to be truly effective, they need to be widely accepted and used by both deaf and mute individuals and the hearing world. This requires a shift in attitudes and a willingness to accommodate new communication methods.

Privacy Concerns: Hand gesture vocalizers collect data about the user's hand movements, which raises privacy concerns. It is important to ensure that this data is collected, stored, and used responsibly to protect the user's privacy.

Technical Challenges: Developing and implementing hand gesture vocalizers involves several technical challenges. These challenges include Optimizing sensor placement, Enhancing gesture recognition algorithms, Improving text-to-speech synthesis, Addressing language barriers.

2.4 “Implementation of Virtual Assistant with Sign Language using Deep Learning and Tensorflow” Valsaraj Vani. [13]

Gesture means it is form of body motions mainly hand, leg etc. Deaf and dumb community perform sign language. This community use sign language for their communication when broadcasting audio is impossible or typing and writing is difficult but there is the vision possibility. At that sign language is the only way of exchanging information between people. Normally sign language is used by everyone when they do not want to speak but this is only way of communication for deaf and dumb community. Sign languages have the same meaning that of spoken language. Sign languages are of two types they are isolated and continuous sign language. Isolated sign language means single gesture mainly of single words whereas continuous sign language means it is a sequence of gesture having meaningful sentences. In this project, isolated American Sign Language recognition techniques is

performed. In the proposed system for gesture recognition image processing and computer visions are used .Gesture recognition enables computer to understand human action and also act as an interpreter between computer and human. This could provide potential to human to interact naturally with computers without any physical contact of mechanical devices. Concepts of CNN and tensorflow libraries are used.CNN is nothing but deep learning algorithm that is capable of assigning biases and weights to different objects in an images and on basis of same it can differentiate one images from another. It consist of different processing layers of image classification and it is designed with the mean of representing functioning of neurons in human brains. The best part of using tensorflow library is that it is an open source with lot of pre designed model useful in deep learning. Tensorflow in our system helps in training the model using provided dataset. It is help in classifying and identifying real time hand gesture help in processing more information a spot more patterns. The most basic explanation of workflow of the system goes as follows - A hand gesture is performed in front of the webcam. This sign gesture is converted to text and the text output is converted to audio and is served as an input to the assistant. The assistant processes the question and responds in audio format. This audio format is converted to text output. The text output will be then displayed on the display screen. The very first step is providing the raw material or dataset. Once the extraction of frame done from the raw dataset the next phase is preprocessing. This is done by key frame extraction using gradient method and after preprocessing next comes the feature extraction is done. Data preprocessing is used to convert the raw data into clear dataset. Key frame extraction is powerful tool that is implements video content by detecting a set of summary key frames to represent video sequences. Features are extracted from the dataset using key gradient based key frame extraction technique these key frames are useful for splitting continuous languages gestures into sequences for sign as for removing uninformative frames. After splitting of gestures features of preprocessed gestures are extracted using orientation histogram is done. The last one is training that is recognition is done using distance techniques finally output will be formed. Objective score is that summarizes the relative difference between two objects in a problem domain. Basically the model will be trained more number of times for the same set of labels, higher is the success rate. But it is required to keep a note that any changes made with the labels folder before training will lead to the system that is being trained for the very first time. In simple terms if it is required to make any changes in the labels folder that is adding new labels or replacing the existing labels, the model will need to be trained again from the beginning. It was observed that for training of around 15 labels on an average

configured system, it takes about 12-15 hours straight of model training for the first time. However retraining of same set of labels requires comparatively lesser amount of time. After training finally the output will be formed in the form of text and make use of that text to speak out. The entire process can be repeated multiple number of times. The hand gesture will be only recognized if they meet the trained dataset standard.

The designed system was successfully able to capture Hand Gestures using the integrated Web Camera and process and convert into text format and display it onto the Input frame and then converted into Audio format. The audio output was being successfully converted into Text format and displayed on the screen. Basically the system can be considered as a boon to people with hearing disabilities or speaking disabilities or both at the same time. These systems would not only bring technology into their Personal lives but also give rise of opportunities in their professional life.

Some potential drawbacks and challenges to consider are;

Accuracy: Hand gesture recognition, especially in a real-world environment, can be challenging. Factors like lighting conditions, background noise, and variations in hand movements may affect the accuracy of the recognition system.

Variability in Sign: Sign languages can vary significantly between regions and even among individuals. Creating a system that accurately recognizes a wide range of sign language gestures may require a large and diverse dataset for training.

Training and Adaptation: Users may need to undergo training to ensure the system accurately recognizes their specific signing style. Adapting the system to individual users' unique gestures can be complex.

Real-Time Processing: Achieving real-time processing, especially on resource-constrained devices, can be challenging. Delays in gesture recognition may hinder the user experience.

Limited Vocabulary: Building a system that recognizes a comprehensive vocabulary in sign language can be resource-intensive. It might be challenging to cover the entire lexicon of a sign language.

UI Complexity: Ensuring a user-friendly interface that accommodates individuals with varying degrees of technical proficiency and sign language skill may be challenging. The user interface should be intuitive and easy to use.

Integration: Integrating the speak engine with a virtual assistant and ensuring coherent and contextually relevant responses may require additional natural language processing (NLP) capabilities, which can be complex.

Privacy and Ethical Considerations: Consider the privacy implications of capturing and processing hand gestures. Ensure that user data is handled securely and that users are informed about how their data will be used.

Hardware Requirements: The system may have specific hardware requirements for optimal performance. Ensure that it is accessible to users with varying levels of technological resources.

Cultural Sensitivity: Consider the cultural aspects of sign language and ensure that the system is designed to respect and represent diverse sign language communities accurately.

CHAPTER 3

SYSTEM OVERVIEW

3.1 Voice and Gesture based Virtual Desktop Assistant

3.1.1 Work Flow

In Fig. 3.1, the intended work flow is depicted by a block diagram. The speech and gesture-based virtual assistants are connect by using the GUI interface that Tkinter creates. The user can select between a gesture-based assistant and a voice assistant when the software launches by constructing two buttons on the GUI window. The window stays open, until the user quits or exits. The speech or gesture-based assistant, on the other hand, is available for use at any time. Users can speak via a microphone to communicate with a voice-based virtual assistant. On webcam, hand motions are captured by a gesture-based virtual assistant. Live video streaming from the webcam is also possible with finger-based mouse pointer control. Following the extraction of commands, the speech input is transformed to text. Similar to this, with the gesture-based assistant, the matching command is retrieved based on the hand gesture after it has been initially recognized by a trained model. Lastly, work is completed, and the helper reads or displays the results. The libraries used for interface and other operations are shown in Fig. 3.2 [4], [6].

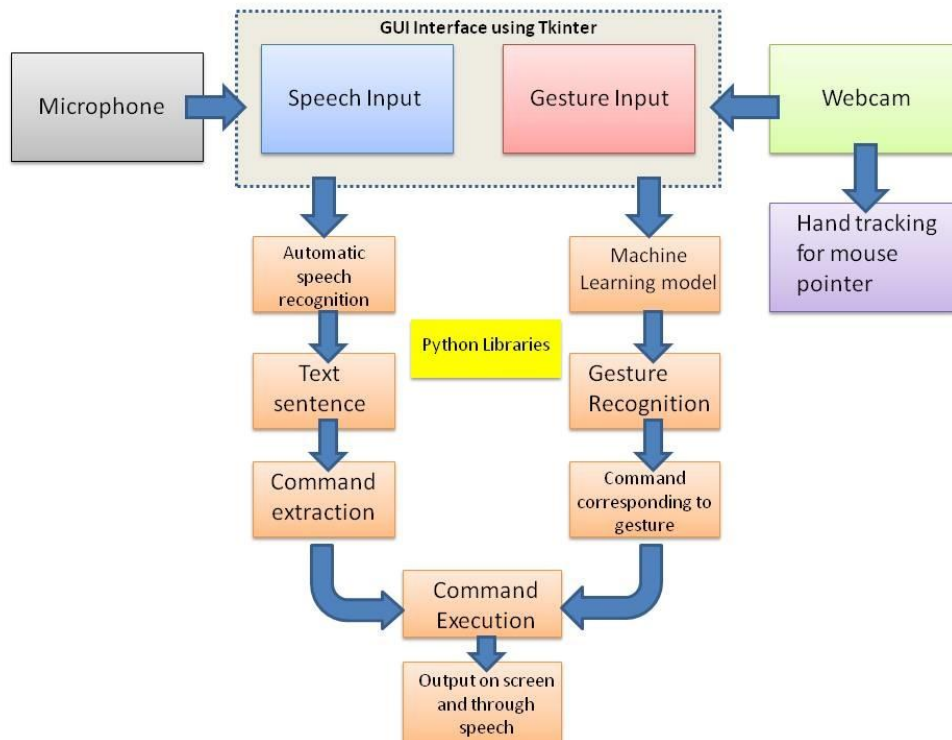


Fig 3.1. Virtual Assistant Block Diagram.

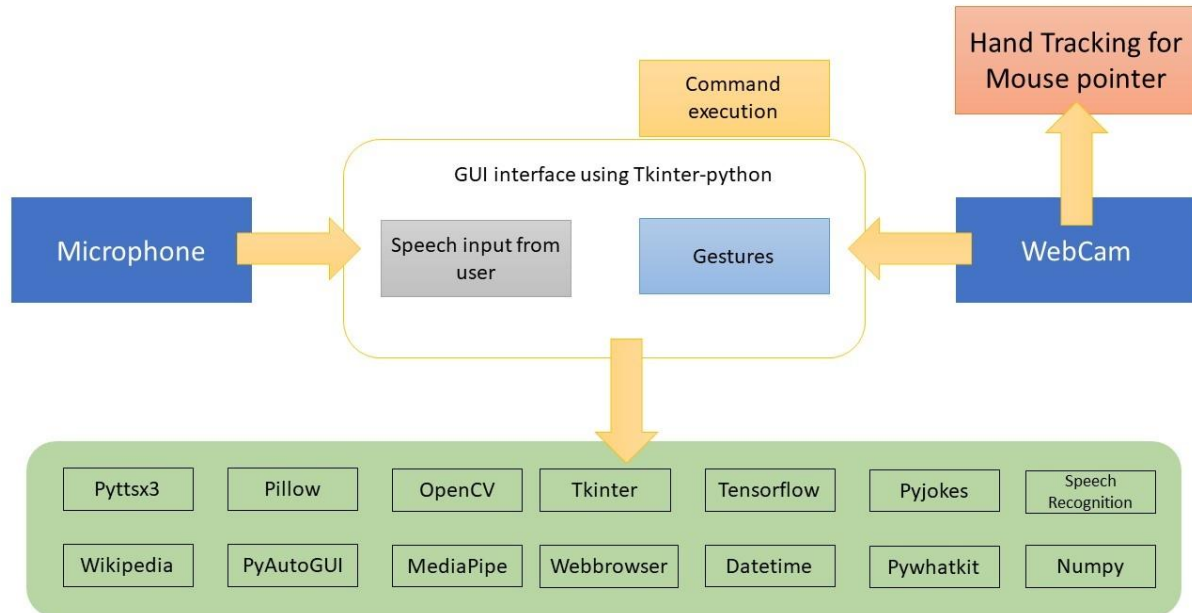


Fig 3.2. Flow Diagram with Various Libraries.

The GUI window is created with two buttons to serve as an interface between a gesture- and speech-based virtual assistant using the Tkinter Python toolkit. Two buttons that give the user access are one for the voice-based virtual assistant and another for the gesture-based virtual assistant. Constructing this using button-related commands. In addition to buttons, widgets are also provided for making listboxes, canvas, radio buttons, labels, text boxes for entry, scrollbars, and many more.

3.1.2 Voice and Gesture Recognition Module

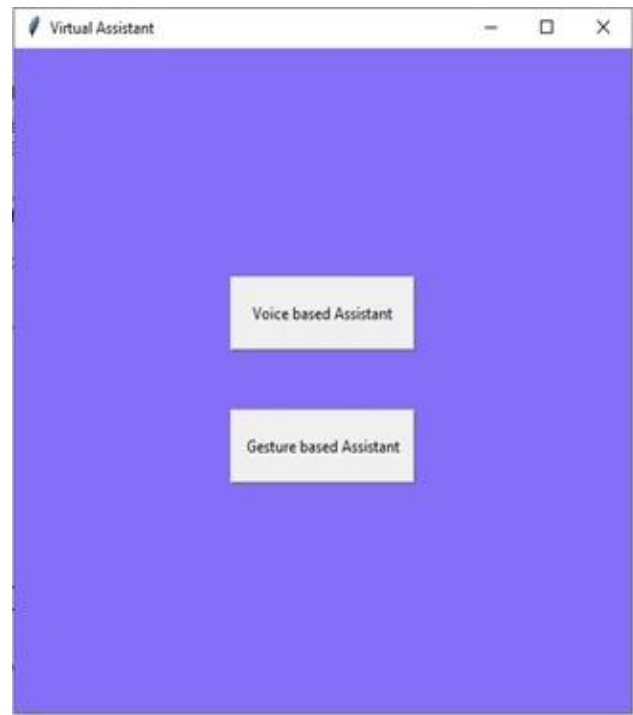
Voice-based Virtual Assistant: Selecting this option launches the voice-based assistant's main function call, named as "VAcall." The model can wish the user and recognize their vocal input to turn speech to text using the two functions found in "VAcall." This is made by using the Python module "speech-recognition," which is designed to identify and translate spoken language to written form.

Gesture-based Virtual Assistant: Upon pressing the gesture-based assistant button, the main user function "HGcall" for the assistant is launched, loading a file containing several named gestures that have been trained for various purposes. When the button is pressed, a window that captures a live webcam feed appears. Every frame is continuously recorded, and the associated action is carried out. The hand detection model operates on the image returned by the palm detection model to return 3D hand key points. The MediaPipe library comprises numerous models, including the palm detection model that returns a box containing only the hand from the full image. To join the knuckles, the lines are drawn between the two objects that is used for the hands [6], [11], [15].

3.1.3 Recognition and Interpretation

In Voice-based Virtual Assistant, the digitally sampled signal is converted to the frequency domain for spectrogram analysis, the fast Fourier transform is applied in the inner workings of this library once the analog voice input has been converted to digital form. Each phoneme has a distinct frequency, and an Artificial Neural Network (ANN) and the Hidden Markov Model are used to randomly arrange these phonemes in the correct order to create text. The tasks associated with the words are completed after comparing the words in this text with the list of words provided in the code.

Objects were developed for text-to-speech software Pyttsx3, a speech recognizer, and a user-created class for user name. Previously mentioned function call is used which is created to record voice data from users once the objects have been initialized. In order to make it easier to retrieve the command from this string, this function converts the voice data that is returned using a speech recognizer object that listens to the users. [2], [3], [5], and [8]. Fig 3.3, shows the commands for the voice assistant demonstration. The developed GUI window, which has two buttons, is shown in Fig. 3.4.

**Fig 3.3.** Voice Assistant Demonstration**Fig 3.4.** GUI based Window

This voice input is passed as a parameter to the other function that was developed for the response, and it is used to perform various tasks like opening Gmail, Instagram, Twitter, displaying a schedule, displaying the weather, playing videos on YouTube, tossing, playing games, performing simple calculations, capturing the current screen, telling random jokes, and searching Wikipedia. The final response from our voice assistant uses a function that uses the object created for pyttsx3 and converts the given text to speech and the virtual assistant speaks it.

In Gesture-based Virtual Assistant, video frames are captured, OpenCV is utilized. MediaPipe is then used to process each frame. The camera displays the landmarks of the gesture through the execution of two for loops. The trained model with a few hand gesture images is given with these landmarks, and it uses OpenCV to predict the corresponding gesture and display its name on the video frame. Every gesture can be recorded with OpenCV, including screen captures, note-taking, email opening, displaying a schedule, displaying the current weather forecast, throwing away outputs, searching on Google, and viewing the news [8], [13], [16], and [17].

Working of the mouse pointer: When the program launches, a webcam is used to record video. Object frames are fed into the system via the OpenCV Python computer vision library. With this system of virtual mouse. The gesture commands for capture, weather, toss, news, Gmail, notes, time, and schedule, among other things, are shown in Figs. 3.5 and 3.6.



Fig 3.5. Gesture Command for Capture, Weather, Toss, News

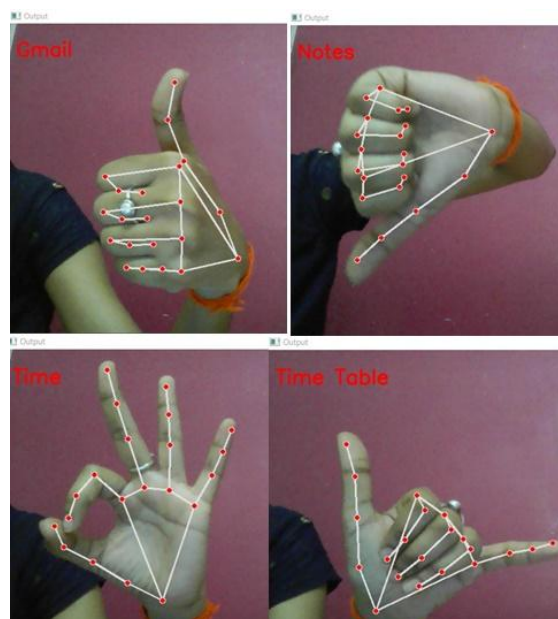


Fig 3.6. Gesture Command for Gmail, Notes, Time, Time Table

Frame by frame each frame is converted from BGR to RGB. Autopy transfers the webcam's fingertip coordinates to the computer window's full screen so that the mouse can control it. Landmarks on the hand are identified using OpenCV and Medipipe models derived from the photos. The mouse pointer can be moved around with the index finger, selected with the thumb, and all fingers raised when there is no action. Thus, a variety of mouse operations can be carried out with hand gestures[7],[9], and [10]. The different gestures for mouse click, pointer movement, and no action are shown in Figs. 3.7, 3.8, and 3.9. [12], [14]. Most existing personal assistants are voice based.

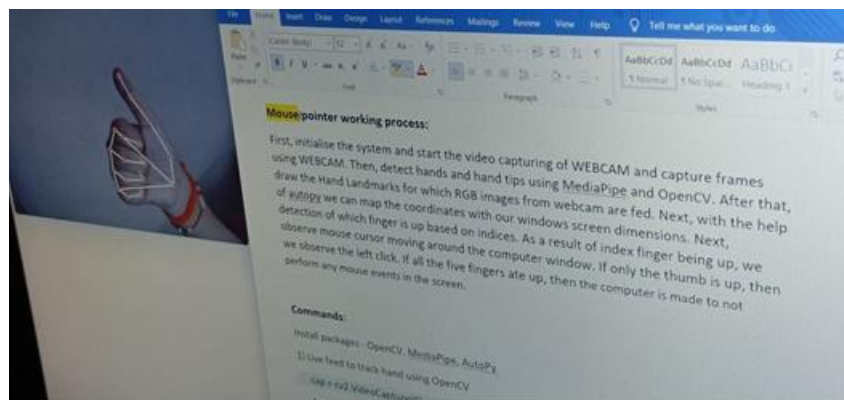


Fig 3.7. Gesture for Computer to Perform Left Click

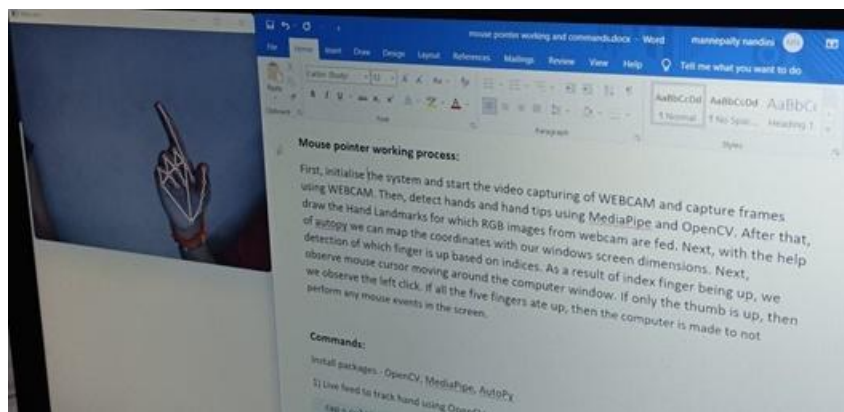


Fig 3.8. Mouse Cursor Moving Around the Computer Window

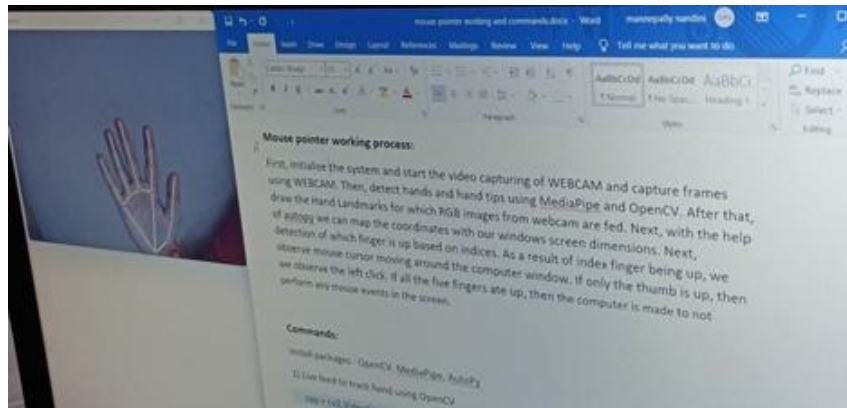


Fig 3.9. Gesture for Computer to Perform No Action

In this system, both a virtual mouse pointer and a gesture-based assistant are integrated. The limitations of current hand gesture recognition techniques include the requirement for stable objects in the frame and a simple background with sufficient light. Even in dim lighting, the suggested model gestures can be identified. Accurate recognition is demonstrated when a gesture changes. Even though the system detects multiple hands, it only takes into account the hand that is closest to the screen and more visible.

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages

Enhanced Accessibility: Voice and gesture-based virtual assistants provide an alternative communication method for individuals with disabilities, particularly those with hearing or speech impairments. This technology empowers them to interact with computers and perform common tasks, enhancing their independence and accessibility.

Natural and Intuitive Interaction: The combination of voice and gesture recognition enables a more natural and intuitive interaction with computers, fostering a more seamless user experience. Users can interact with their devices using their natural language and gestures, reducing the need to learn complex keyboard shortcuts or mouse movements.

Multimodal Input Handling: Voice and gesture-based virtual assistants can handle multiple input modalities simultaneously, providing flexibility and convenience. Users can switch between voice and gesture commands seamlessly, depending on their preference or the situation.

Contactless Interaction: Voice and gesture-based virtual assistants offer a contactless interaction method, eliminating the need for physical contact with devices. This is particularly beneficial in situations where hygiene is a concern, such as healthcare settings or during outbreaks like the COVID-19 pandemic.

Enhanced Productivity: Voice and gesture-based virtual assistants can improve productivity by allowing users to multitask more effectively. Users can perform tasks hands-free while engaging in other activities, such as cooking, exercising, or commuting.

Personalized Interaction: Voice and gesture-based virtual assistants can be personalized to adapt to individual user preferences, speech patterns, and gesture styles. This personalization enhances the user experience and makes the interaction more natural and intuitive.

Context-Aware Interaction: Voice and gesture-based virtual assistants can incorporate context awareness to understand the user's intent and provide more relevant responses. This context awareness can improve the effectiveness of search queries, task execution, and personalized recommendations.

Ubiquitous Presence: Voice and gesture-based virtual assistants have the potential to become ubiquitous, embedded in homes, cars, offices, and public spaces. This widespread presence

will provide users with a convenient and hands-free interface for various tasks, seamlessly integrated into their daily lives.

4.2 Disadvantages

Limited Recognition Accuracy: Voice and gesture recognition technologies are still under development, and their accuracy can be affected by various factors, such as background noise, lighting conditions, and individual accents or gestures. This can lead to misinterpretations and frustration for users.

Privacy Concerns: Voice and gesture-based virtual assistants collect and process a significant amount of user data, including speech recordings and gesture patterns. This raises privacy concerns, as there is a risk of this data being misused or accessed without consent.

Security Vulnerabilities: Voice and gesture-based virtual assistants are susceptible to security vulnerabilities, such as voice or gesture spoofing attacks. These attacks can be used to trick the system into responding to unauthorized commands or accessing sensitive information.

Limited Multitasking Capabilities: While voice and gesture-based virtual assistants can handle multiple input modalities, their ability to handle complex multitasking scenarios is still limited. This can be challenging in situations where users need to perform multiple tasks simultaneously or switch between different tasks frequently.

Potential for Misinterpretation: The natural language processing (NLP) capabilities of voice and gesture-based virtual assistants are evolving, but they are still susceptible to misinterpreting complex or nuanced language. This can lead to misunderstandings and frustration, especially in situations where precise instructions or contextual understanding are critical.

Potential for Social Isolation: Overreliance on voice and gesture-based virtual assistants may lead to social isolation, as it reduces the need for face-to-face interaction and physical touch. This can have negative consequences for social development, empathy, and emotional well-being.

Job Displacement Concerns: The increasing adoption of voice and gesture-based virtual assistants may lead to job displacement in certain sectors, such as customer service or data entry. This raises concerns about the impact on employment and the need for retraining and reskilling programs.

CHAPTER 5

FUTURE SCOPE

The future scope of voice and gesture-based virtual assistants is vast and promising, with the potential to revolutionize the way user interact with computers and technology. Some areas where technology is expected to expand and evolve are,

More sophisticated natural language processing (NLP): It will be incorporated into voice and gesture-based assistants to comprehend complex natural language, enabling more nuanced and context-aware interactions.

Improved Gesture Recognition: With increased accuracy and resilience, gesture recognition will allow users to operate devices and apps with increasingly complex and expressive movements.

Cross-Platform Compatibility: As a greater variety of platforms and devices become compatible with voice and gesture-based assistants, seamless interaction across diverse ecosystems will be possible.

Ubiquitous Presence: Voice and gesture-based assistants will be integrated into homes, vehicles, workplaces, and public areas, making them omnipresent and able to perform a variety of tasks with ease and without requiring the user's hands.

Applications that are Domain-Specific: Voice and gesture-based assistants will be developed with particular industries in mind, like healthcare, education, and entertainment, to offer enhanced user experiences and specialized features.

Multimodal Interaction: Users will be able to transition between various modalities with ease thanks to the integration of voice, gesture, and other multimodal input methods.

Personalization and Adaptation: As users' speech patterns, gestures, and preferences change, voice and gesture-based assistants will become more customized.

AR Integration: Integration of voice and gesture-based assistants with augmented reality (AR) technologies will allow users to interact with virtual objects and environments through voice commands and natural gestures.

Domain-Specific Applications: With specialized functions and improved user experiences, voice and gesture-based assistants will be adapted to particular domains, such as healthcare, education, and entertainment.

Social interaction and emotional intelligence: It will be developed by voice and gesture-based assistants, giving them the ability to identify emotions, react sympathetically, and carry on natural dialogues.

Accessibility and Inclusion: Voice and gesture-based assistants will be essential in advancing accessibility and inclusion because they will give people with disabilities more ways to communicate and improve their tech-interaction skills.

With these developments, voice and gesture-based virtual assistants will become incredibly strong and adaptable instruments that have the potential to drastically change both our everyday lives and how th users engage with the digital world.

CHAPTER 6

CONCLUSION

A voice-activated virtual assistant with hand gestures is designed specifically for individuals with physical disabilities enabling them to complete a variety of tasks on their desktop computers with just their voice and hand gestures. Added a finger-operated mouse pointer which is easy for paralyzed individuals to use their fingers to control the mouse pointer. The apps are helpful for a few specific tasks that can be further enhanced with the help of artificial intelligence. Numerous industries, including robotics, education, healthcare, home automation, automobile automation, and security control can benefit from the use of this suggested application. This system's primary benefit is that it can function even in low visibility situations.

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